

CRUD Operations

Create

```
bookstore> // insert 5 additional books
| db.books.insertMany([
|   { title: "Dune", author: "Frank Herbert", category: "Sci-Fi", price: 24, in_stock: true, published_year: 1965, rating: 4.8 },
|   { title: "The Hobbit", author: "J.R.R. Tolkien", category: "Fantasy", price: 19, in_stock: true, published_year: 1937, rating: 4.9 },
|   { title: "Educated", author: "Tara Westover", category: "Memoir", price: 22, in_stock: false, published_year: 2018, rating: 4.7 },
|   { title: "The Innovator's Dilemma", author: "Clayton Christensen", category: "Business", price: 33, in_stock: true, published_year: 1997, rating: 4.4 },
|   { title: "Structure and Interpretation of Computer Programs", author: "Harold Abelson", category: "Programming", price: 45, in_stock: true, published_year: 1985, rating: 4.7 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a4e08a55a2372fd730cab7e'),
    '1': ObjectId('6a4e08a55a2372fd730cab7f'),
    '2': ObjectId('6a4e08a55a2372fd730cab80'),
    '3': ObjectId('6a4e08a55a2372fd730cab81'),
    '4': ObjectId('6a4e08a55a2372fd730cab82')
  }
}
```

Read

```
bookstore> // Find all books in the "Programming" category
| db.books.find({category: 'Programming'})
[
  {
    _id: ObjectId('6a4e08945a2372fd730cab60'),
    title: 'Clean Code',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 35,
    in_stock: true,
    published_year: 2008,
    rating: 4.8
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab61'),
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    category: 'Programming',
    price: 42,
    in_stock: true,
    published_year: 1999,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab62'),
    title: 'Design Patterns',
    author: 'Erich Gamma',
    category: 'Programming',
    price: 55,
    in_stock: true,
    published_year: 1994,
    rating: 4.6
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab63'),
    title: 'Refactoring',
    author: 'Martin Fowler',
    category: 'Programming',
    price: 48,
    in_stock: false,
    published_year: 2018,
    rating: 4.5
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab64'),
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    category: 'Programming',
    price: 60,
    in_stock: true,
    published_year: 2001,
    rating: 4.9
  }
]
```

```
bookstore> // Find books published after 2015
| db.books.find({published_year: {$gt: 2015}})
[
  {
    _id: ObjectId('6a4e08945a2372fd730cab63'),
    title: 'Refactoring',
    author: 'Martin Fowler',
    category: 'Programming',
    price: 48,
    in_stock: false,
    published_year: 2018,
    rating: 4.5
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab66'),
    title: 'Clean Architecture',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 38,
    in_stock: true,
    published_year: 2017,
    rating: 4.4
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab72'),
    title: 'The Gene',
    author: 'Siddhartha Mukherjee',
    category: 'Science',
    price: 32,
    in_stock: false,
    published_year: 2016,
    rating: 4.6
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab73'),
    title: 'Astrophysics for People in a Hurry',
    author: 'Neil deGrasse Tyson',
    category: 'Science',
    price: 20,
    in_stock: true,
    published_year: 2017,
    rating: 4.4
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab7c'),
    title: 'Atomic Habits',
    author: 'James Clear',
    category: 'Self-Help',
```

```
bookstore> // Find books priced above $40
| db.books.find({price: {$gt: 40}})
[
  {
    _id: ObjectId('6a4e08945a2372fd730cab61'),
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    category: 'Programming',
    price: 42,
    in_stock: true,
    published_year: 1999,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab62'),
    title: 'Design Patterns',
    author: 'Erich Gamma',
    category: 'Programming',
    price: 55,
    in_stock: true,
    published_year: 1994,
    rating: 4.6
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab63'),
    title: 'Refactoring',
    author: 'Martin Fowler',
    category: 'Programming',
    price: 48,
    in_stock: false,
    published_year: 2018,
    rating: 4.5
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab64'),
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    category: 'Programming',
    price: 89,
    in_stock: true,
    published_year: 2009,
    rating: 4.9
  },
  {
    _id: ObjectId('6a4e08a55a2372fd730cab82'),
    title: 'Structure and Interpretation of Computer Programs',
    author: 'Harold Abelson',
```

```
bookstore> // Find books currently in stock
| db.books.find({in_stock: true })
[
  {
    _id: ObjectId('6a4e08945a2372Fd730cab60'),
    title: 'Clean Code',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 35,
    in_stock: true,
    published_year: 2008,
    rating: 4.8
  },
  {
    _id: ObjectId('6a4e08945a2372Fd730cab61'),
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    category: 'Programming',
    price: 42,
    in_stock: true,
    published_year: 1999,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4e08945a2372Fd730cab62'),
    title: 'Design Patterns',
    author: 'Erich Gamma',
    category: 'Programming',
    price: 55,
    in_stock: true,
    published_year: 1994,
    rating: 4.6
  },
  {
    _id: ObjectId('6a4e08945a2372Fd730cab64'),
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    category: 'Programming',
    price: 89,
    in_stock: true,
    published_year: 2009,
    rating: 4.9
  },
  {
    _id: ObjectId('6a4e08945a2372Fd730cab65'),
```

```
bookstore> // Find books written by a specific author
```

```
| db.books.find({author: "George Orwell" })
```

```
[
```

```
{
  _id: ObjectId('6a4e08945a2372fd730cab67'),
  title: '1984',
  author: 'George Orwell',
  category: 'Fiction',
  price: 15,
  in_stock: true,
  published_year: 1949,
  rating: 4.9
},
```

```
{
  _id: ObjectId('6a4e08945a2372fd730cab68'),
  title: 'Animal Farm',
  author: 'George Orwell',
  category: 'Fiction',
  price: 12,
  in_stock: true,
  published_year: 1945,
  rating: 4.7
}
```

```
]

```

```
bookstore> // Find books with a rating greater than 4.5
| db.books.find({rating: {$gt: 4.5}})
[
  {
    _id: ObjectId('6a4e08945a2372fd730cab60'),
    title: 'Clean Code',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 35,
    in_stock: true,
    published_year: 2008,
    rating: 4.8
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab61'),
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    category: 'Programming',
    price: 42,
    in_stock: true,
    published_year: 1999,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab62'),
    title: 'Design Patterns',
    author: 'Erich Gamma',
    category: 'Programming',
    price: 55,
    in_stock: true,
    published_year: 1994,
    rating: 4.6
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab64'),
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    category: 'Programming',
    price: 89,
    in_stock: true,
    published_year: 2009,
    rating: 4.9
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab65'),
    title: 'Code Complete',
    author: 'Steve McConnell',

```

Update

```
bookstore> // Change the price of a book
| db.books.updateOne({ title: 'The Hobbit' }, { $set: { price: 28 } } )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
bookstore> // Update stock availability
| db.books.updateOne({title: 'Educated'}, {$set: {in_stock: true}} )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
bookstore> // Increase the rating of a selected book
| db.books.updateOne({title: 'Educated'}, {$set: {rating: 4.8}} )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Delete

```
bookstore> // Delete at least 2 books from the collection
| db.books.deleteMany({ category: "Business", rating: {$lt: 3.5}} )
{ acknowledged: true, deletedCount: 2 }
```


Aggregation Framework

```
bookstore> // avg book price/category
| db.books.aggregate([
|   {$group: { _id: '$category', avg_price: {$avg: '$price' }} },
|   {$set: { avg_price: {$round: ['$avg_price', 2] }} },
|   {$sort: {avg_price: -1 } }
| ])
[
  { _id: 'Programming', avg_price: 49 },
  { _id: 'Business', avg_price: 28.25 },
  { _id: 'Fantasy', avg_price: 28 },
  { _id: 'Science', avg_price: 26.17 },
  { _id: 'History', avg_price: 25 },
  { _id: 'Sci-Fi', avg_price: 24 },
  { _id: 'Memoir', avg_price: 22 },
  { _id: 'Self-Help', avg_price: 19 },
  { _id: 'Fiction', avg_price: 15 }
]
```

```
bookstore> // #books/category
| db.books.aggregate([
|   {$group: { _id: '$category', tot_books: {$sum: 1} } },
|   {$sort: {tot_books: -1 } }
| ])
[
  { _id: 'Programming', tot_books: 8 },
  { _id: 'Fiction', tot_books: 7 },
  { _id: 'Science', tot_books: 6 },
  { _id: 'Business', tot_books: 4 },
  { _id: 'History', tot_books: 3 },
  { _id: 'Self-Help', tot_books: 2 },
  { _id: 'Sci-Fi', tot_books: 1 },
  { _id: 'Memoir', tot_books: 1 },
  { _id: 'Fantasy', tot_books: 1 }
]
```

```
bookstore> // avg rating/category
| db.books.aggregate([
|   {$group: { _id: '$category', avg_rating: {$avg: '$rating'} } },
|   {$set: { avg_rating: {$round: ['$avg_rating', 2] } } },
|   {$sort: {avg_rating: -1 } }
| ])
[
  { _id: 'Fantasy', avg_rating: 4.9 },
  { _id: 'Sci-Fi', avg_rating: 4.8 },
  { _id: 'Memoir', avg_rating: 4.8 },
  { _id: 'Self-Help', avg_rating: 4.7 },
  { _id: 'Programming', avg_rating: 4.65 },
  { _id: 'Science', avg_rating: 4.62 },
  { _id: 'Fiction', avg_rating: 4.61 },
  { _id: 'History', avg_rating: 4.5 },
  { _id: 'Business', avg_rating: 4.47 }
]
```

```
bookstore> // top-5 expensive books
| db.books.find().sort({ price: -1, _id: 1 }).limit(5)
[
  {
    _id: ObjectId('6a4e08945a2372fd730cab64'),
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
    category: 'Programming',
    price: 89,
    in_stock: true,
    published_year: 2009,
    rating: 4.9
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab62'),
    title: 'Design Patterns',
    author: 'Erich Gamma',
    category: 'Programming',
    price: 55,
    in_stock: true,
    published_year: 1994,
    rating: 4.6
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab63'),
    title: 'Refactoring',
    author: 'Martin Fowler',
    category: 'Programming',
    price: 48,
    in_stock: false,
    published_year: 2018,
    rating: 4.5
  },
  {
    _id: ObjectId('6a4e08a55a2372fd730cab82'),
    title: 'Structure and Interpretation of Computer Programs',
    author: 'Harold Abelson',
    category: 'Programming',
    price: 45,
    in_stock: true,
    published_year: 1985,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4e08945a2372fd730cab61'),
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    category: 'Programming',
    price: 42,
    in_stock: true,
    published_year: 1999,
    rating: 4.7
  }
]
```